INCH-POUND

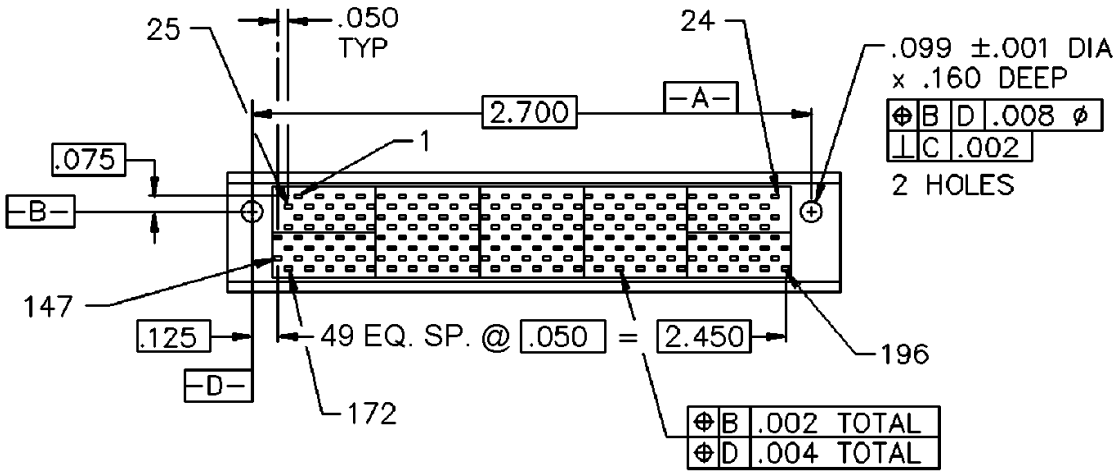
MIL-DTL-32234/4
16 January 2007

DETAIL SPECIFICATION SHEET

CONNECTOR, ELECTRICAL, ULTRA HIGH DENSITY, MODULAR, BLADE AND FORK, DAUGHTER BOARD, EIGHT ROW, 196 CONTACT POSITIONS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-32234.



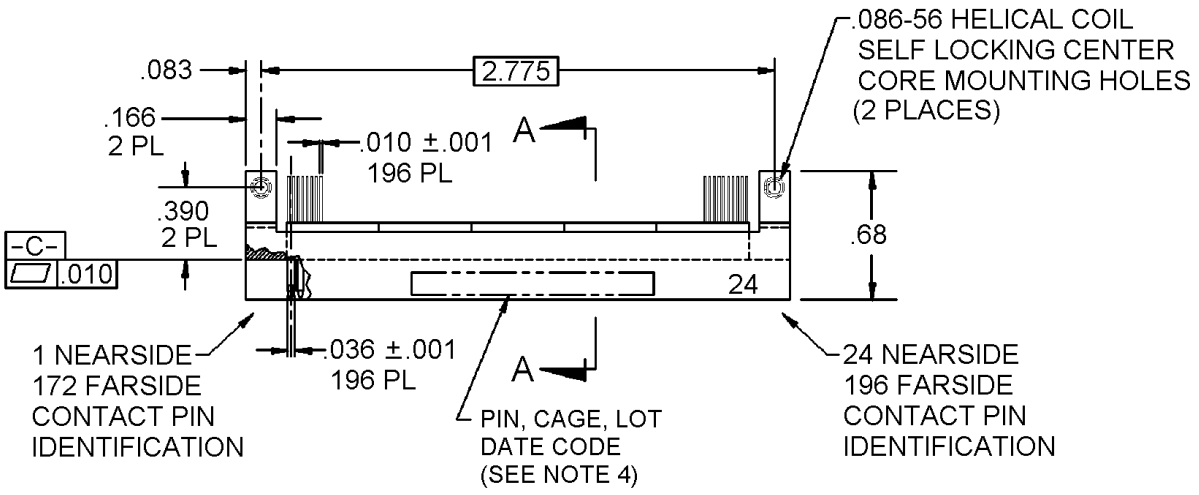
FRONT VIEW

Inches	mm
.001	0.03
.002	0.05
.004	0.10
.008	0.20
.050	1.27
.075	1.91
.099	2.51
.125	3.18
.160	4.06
2.450	62.23
2.700	68.58

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances for 2 place decimals are ±.01 inch (0.3 mm), 3 place decimals are ±.005 inch (0.13 mm).

FIGURE 1. Connector face view.

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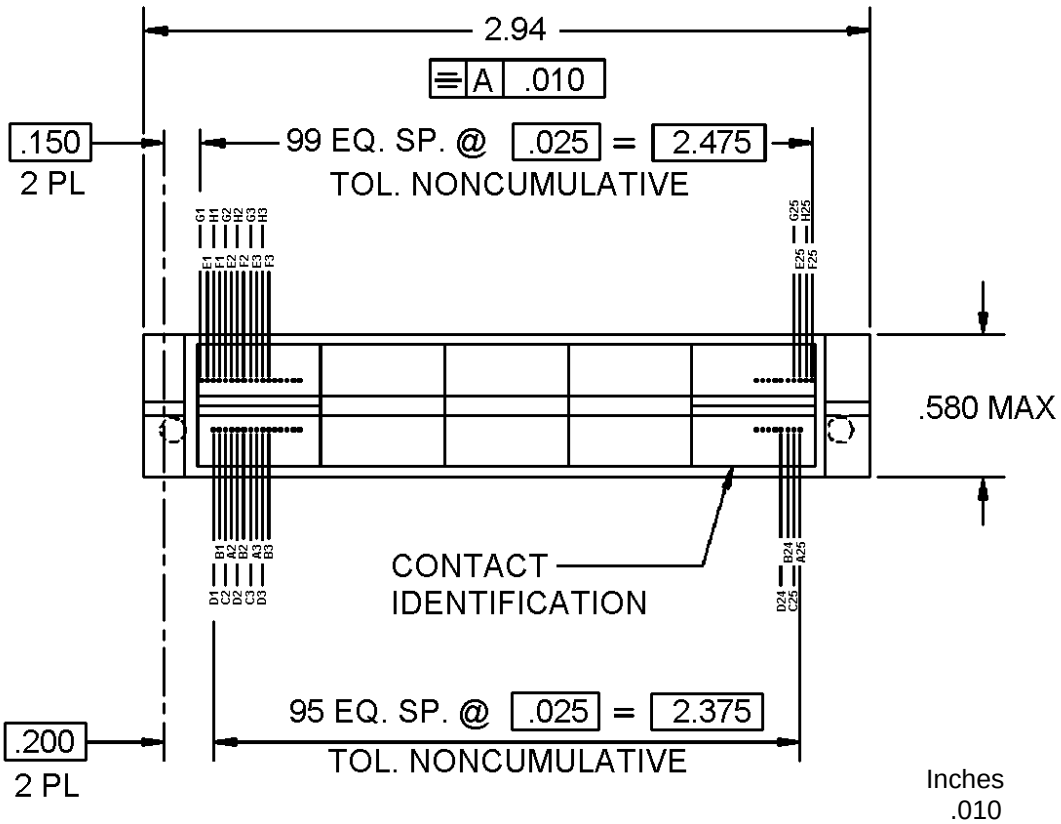
TOP VIEW

Inches	mm
.001	0.03
.010	0.25
.036	0.91
.083	2.11
.086	2.18
.166	4.22
.390	9.91
.68	17.3
2.775	70.49

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances for 2 place decimals are ±.01 inch (0.3 mm), 3 place decimals are ±.005 inch (0.13 mm).
4. Part or Identifying Number (PIN).

FIGURE 2. Connector top view.

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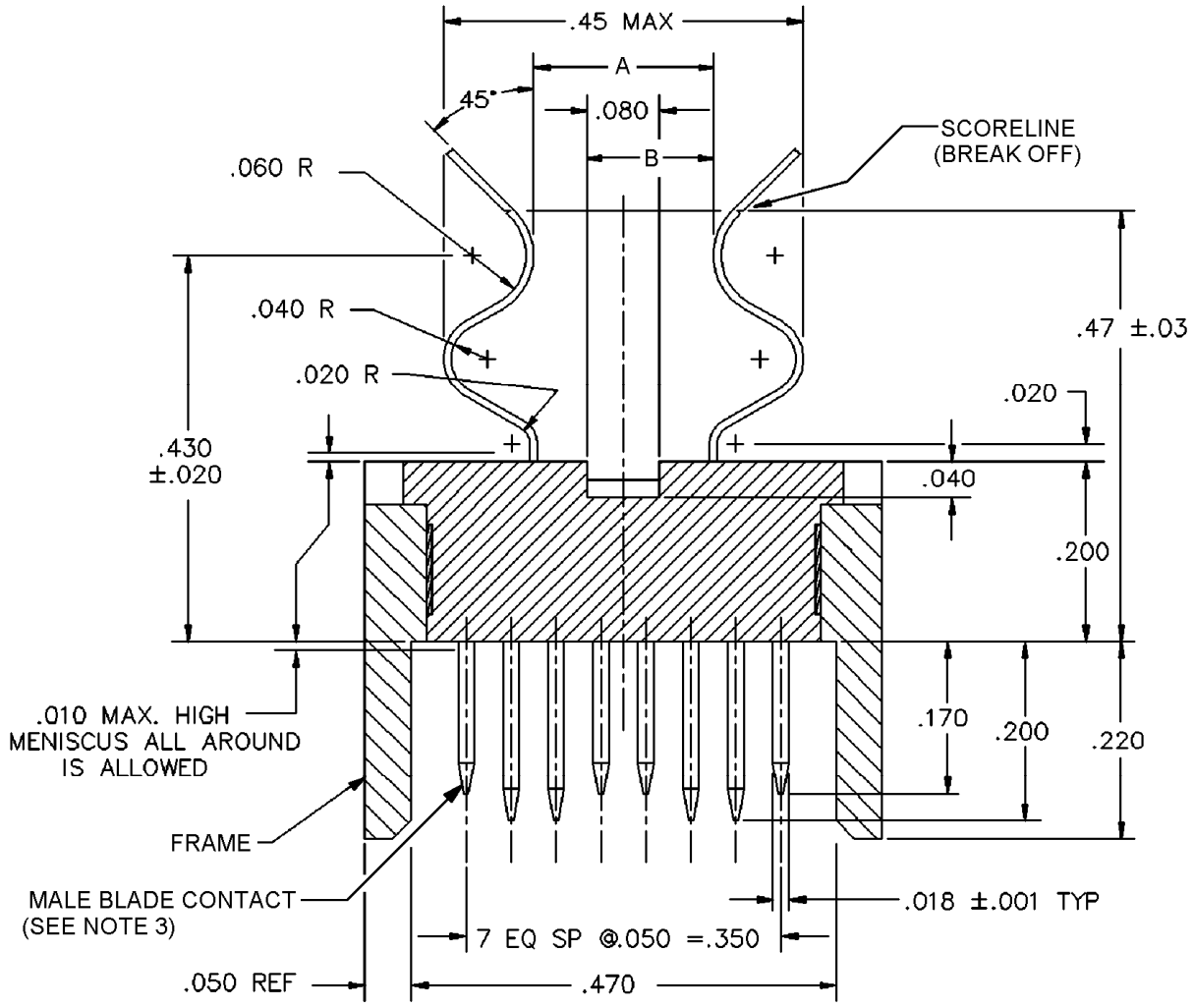


Inches	mm
.010	0.25
.025	0.64
.150	3.81
.200	5.08
.580	14.73
2.375	60.33
2.475	62.87
2.94	74.68

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances for 2 place decimals are ±.01 inch (0.3 mm), 3 place decimals are ±.005 inch (0.13 mm).

FIGURE 3. Connector rear view.

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SECTION A–A

MOUNTING EAR OMITTED FOR CLARITY

FIGURE 4. Connector cross sectional view.

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Inches	mm	Inches	mm
.001	0.03	.170	4.32
.010	0.25	.200	5.08
.018	0.46	.220	5.59
.020	0.51	.350	8.89
.03	0.8	.430	10.92
.040	1.02	.45	11.43
.060	1.52	.47	12.0
.080	2.03	.470	11.94

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Pin lengths may be either 4 rows with a .170 inch (4.32 mm) pin extension and 4 rows with a .200 inch (5.08 mm) pin extension or 2 outside rows at .170 inch (4.32 mm) pin extension and the 6 inside rows at .200 inch (5.08 mm) pin extension.
4. Unless otherwise specified, tolerances for 2 place decimals are ±.01 inch (0.3 mm), 3 place decimals are ±.005 inch (0.13 mm); angular tolerance is ±0°30’.

FIGURE 4. Connector cross sectional view - Continued.

REQUIREMENTS:

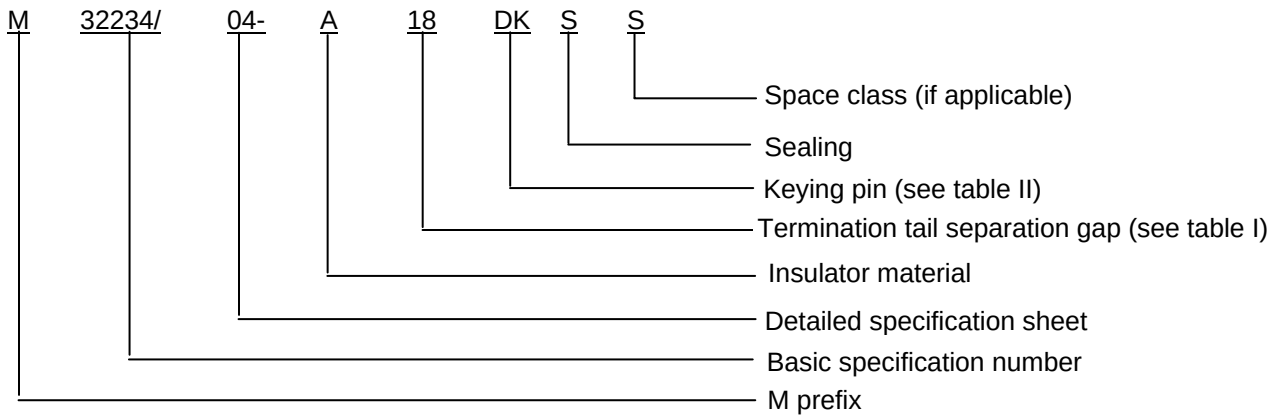
Dimensions and configurations: The design, construction, and physical dimensions shall be as specified on figures 1, 2, 3, 4 and MIL-DTL-32234.

Connector termination. Connectors are surface mount solder termination.

Mating connectors. These connectors mate with backpanel mounted connectors in accordance with MIL-DTL-32234/3.

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PIN example:



Insulator material. Insulator material shall be fabricated using one of the following plastic materials:

A - Thermoplastic, type GST-40F in accordance with MIL-M-24519, or ASTM D4067.
B - Thermoplastic, type GLCP-30 F in accordance with MIL-M-24519, or ASTM D5138.
C - Glass filled diallyl phthalate type SDG-F or GDI-30F in accordance with ASTM D5948.

Termination tail separation (gap). The PIN will include a number as specified in table I, which will identify the connector termination configuration.

TABLE I. Termination tail separation (gap) (see figure 3). 1/ 2/

Tail separation (gap)	Gap A Inches (mm)	B Inches (mm)
53	.053 (1.35) to .083 (2.11)	.084 (2.13)
18	.118 (3.00) to .148 (3.76)	.117 (2.97)
38	.138 (3.51) to .168 (4.27)	.127 (3.23)
40	.140 (3.56) to .170 (4.32)	.129 (3.28)
58	.158 (4.01) to .188 (4.78)	.137 (3.48)
75	.175 (4.44) to .205 (5.21)	.145 (3.68)
84	.184 (4.67) to .214 (5.44)	.149 (3.78)
20	.201 (5.11) to .231 (5.87)	.158 (4.01)

1/ Dimensions are in inches.
2/ Metric equivalents are given for information only.

Keying pins. The PIN will contain a single character for this feature (see table II).

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TABLE II. Keying feature. 1/

Designator	Key or bushing type
DK	“D” keying pin
PK	“PIE” keying pin
DB	“D” keying bushing
PB	“PIE” keying bushing
SP	To be ordered separately (see MIL-DTL-32234/9)

1/ For dimensions and configurations, see MIL-DTL-32234/9.

Sealing. The PIN will contain a single character to indicate if the connector termination area is sealed from the contact blades, which extend above the connector insulator blocks.

N - Not sealed.
S - Sealed or molded in.

Space class.

S - Space class. Thermal vacuum outgassing testing required. OEM is to specify acceptable test specification and outgassing requirements/limits.
Blank - For non-space applications.

Module displacement force. The connector shall allow for cold-wall clamping of attached daughter card modules. The connector shall require 75 pounds (34 Kg) minimum to displace the connector frame .003 inches (0.08 mm) laterally for single-sided mounting ears (non-float mount) and .006 inches (0.15 mm) for double-sided mounting ears (float mount) in either direction when tested in accordance with MIL-DTL-32234. No degradation of performance shall be allowed after this test.

Connector materials.

Material requirements. The materials to be used in construction and fabrication shall be as specified herein. Materials required, but not specified in this slash sheet shall be in accordance with MIL-DTL-32234.

Connector insulator. The connector insulator blocks shall be fabricated from thermoplastic, glass reinforced, in accordance with MIL-M-24519, type GPT-30F.

Referenced documents. In addition to MIL-DTL-32234, this document references the following:

MIL-DTL-32234/3	ASTM D4067
MIL-DTL-32234/9	ASTM D5138
MIL-M-24519	ASTM D5948

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-4829-004)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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